

Q1. (a) Find $P(X > \sqrt{Y})$ if X and Y are jointly distributed with PDF

$$f(x, y) = x + y, \quad 0 \leq x \leq 1, 0 \leq y \leq 1.$$

(b) Find $P(X^2 < Y < X)$ if X and Y are jointly distributed with PDF

$$f(x, y) = 2x, \quad 0 \leq x \leq 1, 0 \leq y \leq 1.$$

Q2. Let X, Y , and Z be independent uniform(0,1) random variables.

(a) Find $P(\frac{X}{Y} \leq t)$ and $P(XY \leq t)$.

(b) Find $P(\frac{XY}{Z} \leq t)$.

Q3. Suppose the distribution of Y , conditional on $X = x$, is $N(x, x^2)$ and the marginal distribution of X is uniform(0,1).

(a) Find $E(X)$, $\text{Var}(Y)$ and $\text{Cov}(X, Y)$.

(b) Show that whether Y/X and X are independent or not.

Q4. Let U be a discrete random variable taking the values $a_1, \dots, a_r$ with probabilities $p_1, \dots, p_r$.

a) Suppose all $a_i \geq 0$, but that $E(U) = 0$. Show then $P(U = 0) = 1$.

b) Suppose that V is a random variable taking the values $b_1, \dots, b_r$ with probabilities $p_1, \dots, p_r$. Show that $\text{Var}(V) = 0$ implies $P(V = E(V)) = 1$.

Q5. In models for the lifetimes of mechanical components one sometimes uses random variables with distribution functions from the so-called Weibull family. Here is an example: $F(x) = 0$ for $x < 0$, and $F(x) = 1 - \exp(-5x^2)$ for $x \geq 0$. Construct a random variable X with this distribution from a $U(0,1)$ random variable.

Q6. We have learned how to generate $Y \sim \text{Exp}(\lambda)$ from $X \sim U(0,1)$. We obtain Z from Y by rounding to the nearest integer greater than Y . What is the distribution of Z ?

Q7. Let X be a continuous random variable. Express the distribution function and probability density of the random variable $Y = -X$ and $Z = 1/X$ in terms of those of X .

Q8. Let X and Y be two independent random variables, where X has an $N(2,5)$ distribution and Y has an $N(5,9)$ distribution. Define $Z = 3X - 2Y + 1$.

(a) Compute $E(Z)$ and $\text{Var}(Z)$.

(b) What is the distribution of Z ?

(c) Compute $P(Z \leq 6)$.